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electronic fan regulators operated by SCR vary the voltage by changing the firing angle of the AC sine wave.

The figure shows the circuit diagram of the simplest fan regulator. This circuit is based on the principle of power control using a triac. It works by varying the firing angle of the triac. Resistors R1 and R2 and capacitor C1 are associated with this. The firing angle can be varied by varying the value of any of these components. Here R1 is selected as the variable element. By varying the value of R1 the firing angle of the triac changes (in simple words, the time duration for which the triac should conduct changes). This directly varies the load power, since the load is driven by the triac. The firing pulses are given to the gate of triac T1 using diac D1.

**Q3.** What's the difference between an op-amp and comparator?

D.K. Saha

Through e-mail

**A3.** Op-amp is short for 'operational amplifier'—a useful little component in many analogue circuits. An ideal op-amp has a positive and a negative input that draw no current, and an output port that can source CV current, where 'V' is equal to the voltage difference between the positive and negative inputs, and 'C' is a very large constant. In regular op-amps, 'C' can vary by up to 50 per cent, but as you'll see, it doesn't matter much.

To use the op-amp as an amplifier, tie the positive input to ground, and stick a resistor R1 between your input and the negative input. Then, tie the output back to the negative port with a resistor R2. The resulting amplification should be equal to  $-R2/R1$ . The best part is that R2 and R1 can usually be substituted with impedance elements like capacitors and inductors, and the equation will still hold after the appropriate Laplace transformations have taken place.

As long as op-amps are used within the spec for frequency and aren't expected to output past their source volt-

ages, they will perform very well.

A voltage comparator is a common tool used in electronic design that signals a change in voltage level. Comparators are used in a wide variety of applications, and can commonly be found in analogue-to-digital converters, digital logic designs and even RAM chips in your computer. The example discussed below is pedagogical, but will suffice. As an example, a simple voltage comparator utilising an operational amplifier serves as an excellent starting point.

A simple op-amp voltage comparator compares two input voltages (V1, V2) and outputs a signal that determines which voltage is greater. This is a very simple task, but it is also very important in electronics.

The principle behind a simple comparator is understanding that op-amps have a gain (A). For most op-amps, 'A' is of the order of 200,000. This value has no units, but can be thought of as a multiplier or constant. The two voltage inputs to the op-amp (V1, V2) are the input voltages to be compared. The output (Vo) is the result of the comparison.

The equation:

$$V_o = A(V_1 - V_2)$$

describes the basic functionality of the op-amp. We will now re-label V2 as  $V_R$  which is short for reference voltage, for the rest of this discussion. This transforms the equation to:

$$V_o = A(V_1 - V_R)$$

As already mentioned, 'A' is very large, so any difference between V1 and  $V_R$  will be greatly magnified (multiplied by a number of the order of 200,000). This causes something called saturation. Op-amps saturate if the output voltage goes higher than the voltage level that actually powers the op-amp.

Answers compiled by Air Cmdr (Retd) P.D. Badoni

Letters and questions for publication may be addressed to Editor, Electronics For You, D-87/1, Okhla Industrial Area, Phase 1, New Delhi 110020 (E-mail: [editsec@efyindia.com](mailto:editsec@efyindia.com)) and should include name and address of the sender.